# Biochemistry Cume
# January 29, 2016

**Total points**: 100 points

1. Mutations are caused by chemical and physical phenomena. Indicate the type of mutation that each of the following reactions or molecules might cause. **[20 (4 x 5) points]**

   **A.** Reactive Oxygen Species (ROS)

   **B.** Alkylating agent

   **C.** Intercalating agents

   **D.** Nitrous acid

2. The *E. coli* chromosome contains 4,640,000 base-pairs. **40 (20 + 20) points**

   **A.** How many turns of the double helix must be unwound during replication of the E.coli chromosome? (Given information: The distance between each base-pair in the *E. coli* chromosome is 3.4 Å, and one-turn of the DNA (comprising the E.coli chromosome) contains 10 base-pairs.)

   **B.** How long would it take to replicate the E.coli chromosome at 37 °C if two replication forks proceed from the origin? Assume the rate of replication to be 1000 base pairs per second.

3. Arthur Kornberg and his colleagues incubated soluble extracts of E.coli with a mixture of dATP, dTTP, dGTP, and dCTP, all labeled with $^{32}P$ in the α-phosphate group. After a time, the incubation mixture was treated with trichloroacetic acid, which precipitates the DNA but not the nucleotide precursors. The precipitate was collected, and the extent of precursor incorporation into DNA was determined from the amount of radioactivity present in the precipitate.

   **A.** If any one of the four nucleotide precursors were omitted from the incubation mixture, would radioactivity be found in the precipitate? Explain. (**15 points**).

   **B.** Would $^{32}P$ be incorporated into the DNA if only dTTP were labeled? Explain (**15 points**).

   **C.** Would radioactivity be found in the precipitate if the β- or γ-phosphate was $^{32}P$ labeled rather than the α-phosphate of the deoxyribonucleotides? Explain (**10 points**).